

Solução do Problema D2Q06C - Solver HiGHS

Análise Computacional

6 de outubro de 2025

1 Informações do Problema

Informações do Problema:

- Nome: D2Q06C
- Número de Variáveis: 5167
- Número de Restrições: 2171
- Inviabilidade Primal: 4.349e-13
- Inviabilidade Dual: 5.335e-13
- Valor Primal: 1.228e+05
- Valor Dual: 1.228e+05
- Gap: 0.000e+00
- Número de Iterações: 34

2 Variáveis Primais e Custos Reduzidos ($x \geq 0$ e $z \geq 0$)

Tabela 1: Variáveis primais (x) e custos reduzidos (z) com valores ≥ 0 do problema D2Q06C

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
3	1.333e+05	1	3.504e-01
4	2.050e+02	2	5.010e-01
6	5.466e+01	10	6.957e-02
7	2.504e+02	11	8.349e-01
8	1.799e+02	12	2.002e-01
9	4.054e+01	13	3.047e-01
36	6.008e+00	14	5.705e-02
52	2.493e+01	17	3.043e-01
56	4.536e+01	19	1.953e-01
58	1.661e+02	20	1.478e-01
68	1.645e+02	21	2.315e-01
89	4.207e+00	22	2.366e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
91	7.571e+00	23	5.117e-01
92	3.860e+00	24	6.086e-01
93	1.134e+01	25	1.889e-01
94	7.216e+00	26	1.728e-01
95	4.823e+00	27	2.193e-01
96	9.000e+00	28	2.084e-01
136	3.924e+03	29	2.604e-01
137	3.924e+03	30	3.048e-01
335	7.321e+00	31	5.532e-01
337	1.136e+01	32	6.866e-01
339	4.119e+01	33	2.125e-01
342	4.707e+01	34	2.384e-01
351	2.824e+02	35	9.341e-02
355	1.176e+02	37	1.362e-01
363	2.364e+01	38	1.074e-01
377	3.194e+00	39	4.802e-01
378	7.647e-01	40	5.896e-01
379	5.548e+00	41	8.567e-02
380	3.225e+01	42	2.769e-02
381	1.789e+01	43	2.121e-01
384	9.304e-01	44	2.313e-01
385	1.810e+01	45	2.259e-01
389	1.982e+01	46	6.718e-01
390	1.661e+00	47	2.404e-01
392	1.060e+01	48	2.730e-01
394	2.118e+01	49	2.497e-01
395	2.385e+01	50	7.051e-01
403	1.181e+02	51	1.451e-02
404	1.109e+02	53	6.320e-02
409	3.468e+02	54	6.167e-01
418	1.096e+02	55	1.290e-01
421	7.285e+00	57	9.422e-02
423	1.104e+02	59	6.434e-02
426	3.727e+02	60	4.163e-01
428	5.000e+02	61	5.229e-01
429	3.597e+01	62	4.266e-01
432	3.776e+02	63	4.909e-01
434	2.433e+02	64	5.626e-01
435	3.977e+01	65	6.648e-01
436	3.043e+02	66	5.792e-01
437	1.261e+01	67	6.405e-01
439	2.288e+02	69	7.945e-02
441	1.319e+01	70	1.656e-02
442	1.059e+01	71	8.090e-02
443	4.750e+01	72	9.776e-02
447	1.363e+02	73	1.697e-01
466	3.168e+01	74	1.164e-01
467	2.937e+01	75	1.777e-01
468	6.553e-01	76	2.626e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
469	5.830e+00	77	2.353e-01
473	1.119e+01	78	6.522e-01
474	3.303e+01	79	6.169e-01
481	1.765e+01	80	2.078e-01
482	6.203e+01	81	2.007e-01
484	3.523e+01	82	9.316e-01
485	2.321e+01	83	8.577e+00
487	8.703e+00	84	8.577e+00
489	1.324e+01	85	3.172e+00
490	2.475e+03	86	8.611e+00
491	6.000e+01	87	1.411e+01
492	8.710e+01	88	9.979e+00
493	1.082e+01	90	1.118e+01
495	1.850e+01	97	1.118e+01
496	6.306e+00	98	1.859e+00
497	7.066e+00	99	1.134e+01
498	4.622e+00	100	1.260e+01
501	3.042e+01	101	4.252e+00
509	7.775e+00	102	5.605e+00
516	3.451e+01	103	4.742e+00
522	1.189e+01	104	4.628e+00
523	8.101e+01	105	1.082e+01
536	8.289e+01	106	6.599e+00
539	1.594e+02	107	1.492e+01
540	1.395e+01	108	9.098e+00
547	3.031e+02	109	1.245e+01
549	1.377e+02	110	1.278e+01
554	3.210e+02	111	1.019e+01
558	1.152e+03	112	1.258e+01
561	2.440e+01	113	1.192e+01
562	2.528e+01	115	6.773e+00
565	3.700e+01	116	1.433e+01
569	5.058e+01	117	1.965e+01
571	1.422e+02	118	5.217e+00
572	9.151e+01	119	1.414e+01
573	1.314e+02	120	4.628e+00
574	1.500e+01	121	8.487e+00
576	3.700e+01	122	1.470e+01
578	2.213e+02	123	4.757e+00
582	3.988e+00	125	1.308e+01
583	1.036e+00	126	1.146e+01
587	5.451e+00	127	9.121e+00
594	5.485e+01	128	3.707e+00
599	7.684e+01	129	9.257e+00
600	2.020e+01	130	6.899e+00
601	2.248e+01	131	7.215e-01
607	2.520e+01	132	3.800e+00
615	1.935e+02	133	5.948e+00
616	5.340e+01	134	3.756e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
620	5.973e+01	135	3.870e+00
624	4.053e+01	138	3.673e-02
628	1.238e+02	139	8.597e+00
629	1.353e+02	140	1.197e-01
630	7.912e+01	141	1.134e+01
633	3.154e+01	142	1.253e+01
636	1.250e+02	143	1.250e+01
637	2.217e+02	144	1.095e+01
638	6.261e+01	145	1.334e+01
639	1.277e+02	146	6.969e+00
640	7.920e+01	147	1.292e+01
649	9.100e+00	148	1.862e+00
651	2.680e-01	149	3.172e+00
652	7.344e+00	150	1.031e+01
653	7.400e+00	151	9.926e+00
655	2.843e+01	152	1.066e+01
659	1.639e+00	153	1.082e+01
660	8.201e+00	154	1.021e+01
662	2.131e+01	155	1.073e+01
663	3.559e+00	156	1.075e+01
664	1.969e+01	157	1.102e+01
665	9.371e+00	158	1.036e+01
668	3.424e+00	159	9.699e+00
669	2.581e+01	160	1.052e+01
670	2.535e+01	161	1.101e+01
671	1.464e+00	162	1.114e+01
672	8.616e+00	163	1.057e+01
674	1.614e+00	164	4.134e+00
675	1.942e+01	165	9.519e+00
677	4.353e+01	166	5.256e+00
678	1.063e+02	167	1.032e+01
679	7.123e+01	168	1.062e+01
680	3.422e+01	169	4.826e+00
681	4.228e+01	170	1.069e+01
682	9.709e+00	171	1.068e+01
685	3.432e+01	172	3.901e+00
686	6.111e+00	173	5.024e+00
687	7.686e+00	174	4.671e+00
690	1.433e+01	175	1.132e+01
693	3.037e+01	176	1.024e+01
696	2.104e+01	177	6.570e+00
698	5.810e+01	178	1.029e+01
708	2.770e+01	179	9.859e+00
711	2.240e+02	180	1.074e+01
713	8.000e+01	181	1.055e+01
715	3.590e+01	182	9.363e+00
717	1.872e+02	183	1.017e+01
718	1.499e+01	184	1.091e+01
721	1.354e+01	185	1.105e+01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
722	5.255e+01	186	4.100e+00
723	5.140e+00	187	9.422e+00
725	9.730e+01	188	1.065e+01
726	2.165e+02	189	1.061e+01
728	3.186e+01	190	4.729e+00
729	1.263e+02	191	1.056e+01
731	1.922e+02	192	1.132e+01
735	6.375e+01	193	1.103e+01
737	6.922e+01	194	1.115e+01
738	1.647e+01	195	1.163e+01
739	1.030e+02	196	1.069e+01
741	8.235e+01	197	1.056e+01
742	2.898e+02	198	1.029e+01
748	3.196e+01	199	9.812e+00
749	5.095e+00	200	1.079e+01
751	5.766e+01	201	1.009e+01
753	4.395e+01	202	1.063e+01
755	7.114e+00	203	1.090e+01
760	3.420e+01	204	1.061e+01
761	5.074e+00	205	9.255e+00
762	1.929e+02	206	1.051e+01
764	2.746e+02	207	1.092e+01
768	1.012e+02	208	1.102e+01
772	2.255e+02	209	1.056e+01
774	4.649e+02	210	4.112e+00
780	3.134e+01	211	1.003e+01
782	1.580e+01	212	5.231e+00
785	6.786e-01	213	9.776e+00
786	7.264e+00	214	1.051e+01
787	1.292e+00	215	4.654e+00
789	4.570e+00	216	1.056e+01
790	4.975e+01	217	8.736e+00
793	4.287e+00	218	9.487e+00
795	4.606e+00	219	1.023e+01
797	3.919e+01	220	9.940e+00
799	1.854e+01	221	9.788e+00
800	2.176e+01	222	9.833e+00
801	1.387e+01	223	9.832e+00
802	1.862e+00	224	9.292e+00
804	1.738e+01	225	9.441e+00
806	1.796e+01	226	9.404e+00
810	6.432e+00	228	1.116e+01
811	3.073e+01	229	1.061e+01
812	4.921e+00	230	1.111e+01
816	2.805e+01	231	1.105e+01
817	3.632e+00	232	1.090e+01
818	1.844e+01	233	5.030e+00
819	1.951e+00	234	1.026e+01
820	5.967e+01	235	1.148e+01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
821	5.595e+01	236	1.046e+01
824	3.055e+01	237	1.028e+01
827	4.352e+01	238	1.039e+01
828	2.023e+01	239	1.032e+01
838	3.480e+01	240	1.024e+01
839	7.839e+01	241	9.933e+00
841	1.571e+01	242	1.023e+01
845	1.533e+01	243	1.017e+01
847	6.667e+01	244	1.054e+01
848	1.258e+01	245	1.055e+01
852	2.000e+02	246	1.052e+01
855	7.686e+01	247	1.022e+01
859	4.696e+02	248	1.048e+01
862	1.922e+01	249	1.037e+01
864	6.600e+00	250	1.051e+01
865	1.075e+02	251	1.045e+01
867	2.577e+02	252	1.102e+01
872	5.254e+01	253	1.071e+01
876	1.215e+02	254	1.032e+01
878	1.999e+01	255	3.679e+00
880	7.334e+01	256	1.057e+01
881	1.312e+01	257	4.690e+00
882	1.308e+02	258	1.019e+01
884	1.080e+02	259	1.101e+01
885	2.955e+02	260	1.047e+01
888	1.653e+02	261	1.107e+01
890	4.841e+01	262	1.023e+01
891	7.174e+01	263	1.045e+01
892	2.671e+01	264	9.747e+00
894	4.550e+01	265	1.041e+01
898	2.531e+01	266	1.037e+01
901	2.851e+02	267	1.047e+01
902	5.987e+01	268	1.045e+01
903	4.970e+01	269	1.078e+01
905	4.841e+00	270	1.070e+01
912	6.510e+00	271	1.009e+01
913	5.677e+01	272	9.269e+00
915	3.383e+01	273	9.799e+00
917	3.942e+00	274	9.796e+00
920	1.644e+00	275	9.841e+00
921	3.673e+00	276	1.032e+01
922	2.269e+01	278	1.088e+01
923	1.077e-01	279	4.058e+00
924	1.067e+01	280	9.609e+00
925	9.302e+00	281	1.075e+01
926	2.402e+01	282	5.059e+00
928	2.789e+01	283	4.619e+00
930	5.872e+01	284	3.854e+00
931	1.939e+01	285	3.635e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
932	8.628e+00	286	4.249e+00
933	2.877e+01	287	1.088e+01
936	5.103e-01	288	1.090e+01
937	5.386e+00	289	1.083e+01
938	1.293e+01	290	1.011e+01
942	5.465e+00	291	1.065e+01
943	2.567e+01	292	1.061e+01
945	7.994e+00	293	1.116e+01
949	3.549e+01	294	1.153e+01
957	3.339e+00	295	1.110e+01
963	3.196e+01	296	1.149e+01
969	2.544e+01	297	1.120e+01
972	2.679e+02	298	1.146e+01
973	1.330e+01	299	1.064e+01
974	1.891e+02	300	1.002e+01
976	4.866e+01	301	1.109e+01
978	7.257e+01	302	1.061e+01
981	1.256e+01	303	1.055e+01
983	1.101e+02	304	1.116e+01
987	2.865e+02	305	1.110e+01
989	1.528e+02	306	1.047e+01
990	2.290e+02	307	1.002e+01
991	1.261e+02	308	1.140e+01
993	3.291e+01	309	1.079e+01
994	2.350e+01	310	4.148e+00
995	2.082e+01	311	1.084e+01
997	3.332e+02	312	1.101e+01
1000	4.525e+01	313	5.083e+00
1001	1.418e+00	314	1.092e+01
1004	2.100e+02	315	8.936e+00
1005	2.350e+02	316	9.603e+00
1008	2.887e-01	317	1.039e+01
1009	1.345e+01	318	1.049e+01
1015	8.265e+00	319	1.059e+01
1016	1.191e+01	320	4.711e+00
1017	2.213e+01	321	1.059e+01
1020	2.214e+01	322	1.320e+01
1024	1.893e+00	323	1.113e+01
1027	4.500e+00	324	1.017e+01
1028	2.099e+01	325	9.990e+00
1029	6.987e+01	326	1.063e+01
1031	5.974e-01	327	9.951e+00
1032	3.431e+00	328	9.646e+00
1033	8.580e-01	329	9.764e+00
1034	1.388e+00	330	1.019e+01
1036	2.237e+01	331	1.030e+01
1038	9.123e-01	332	1.044e+01
1039	1.469e+01	333	1.141e+01
1040	2.694e+00	334	3.155e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1041	2.101e+01	336	2.267e-02
1043	5.304e+01	338	3.550e-01
1044	5.880e+01	340	6.244e-01
1045	6.234e+01	341	3.334e-03
1047	4.953e+01	343	1.151e-01
1049	1.252e+01	344	1.544e-01
1050	8.906e-01	345	1.208e+00
1054	3.592e+01	346	8.531e-01
1056	5.570e+00	347	1.008e-01
1058	3.591e+00	348	1.767e-01
1060	6.136e+01	349	1.577e+00
1071	1.004e+02	352	1.004e-01
1072	3.939e+02	353	1.589e+00
1073	4.200e+01	354	1.112e-05
1077	4.494e+02	356	1.578e-04
1078	4.742e+01	357	9.195e-02
1080	1.630e+02	358	1.043e+00
1082	1.145e+02	359	8.357e-03
1086	9.939e+01	360	3.360e-01
1087	3.900e+02	361	3.360e-01
1088	4.158e+01	362	3.360e-01
1100	8.606e+01	364	8.798e-01
1101	3.053e+01	365	8.574e-01
1103	2.131e+01	366	2.143e+01
1104	1.795e+01	367	2.134e+01
1105	1.253e+02	368	2.148e+01
1106	2.655e+02	369	1.866e+01
1107	3.220e+01	370	1.339e+01
1113	5.925e+00	371	1.421e+01
1114	2.486e+02	372	6.897e+00
1116	8.740e+01	375	2.055e+00
1118	1.668e+02	376	2.184e+00
1122	4.082e+00	382	1.768e-01
1124	3.181e+02	383	1.976e-01
1125	3.785e+02	386	3.991e+00
1127	2.727e+02	387	2.623e+00
1130	2.218e+01	388	1.200e+01
1131	3.501e+02	391	9.784e+00
1135	1.888e+02	393	1.379e+00
1141	1.326e+02	399	8.681e+00
1146	2.875e+01	400	3.326e-01
1148	2.973e+01	401	2.550e-01
1149	2.732e+01	402	2.352e-01
1154	3.363e+01	405	1.035e+01
1155	9.087e+01	406	9.414e+00
1156	9.707e+01	407	9.388e+00
1157	1.296e+02	408	1.323e+00
1159	1.303e+01	412	1.151e-01
1160	7.000e+00	413	1.545e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1161	1.805e+01	414	1.208e+00
1165	1.394e+01	415	1.008e-01
1167	4.102e+01	416	1.767e-01
1172	4.899e-01	417	1.577e+00
1173	2.319e+01	419	1.004e-01
1175	6.656e+01	420	1.589e+00
1176	1.508e+01	422	4.734e-05
1177	1.021e+02	424	9.180e-02
1179	1.771e+01	425	1.042e+00
1180	6.313e+00	427	8.357e-03
1182	2.029e-03	430	1.524e+01
1184	1.301e+01	431	1.134e+01
1185	2.821e+01	433	2.836e-02
1186	2.735e+01	438	3.629e-14
1190	3.588e+03	444	1.520e-01
1191	5.242e+01	445	1.202e-01
1192	4.796e+00	446	1.731e-01
1194	1.184e+01	448	5.576e-02
1195	2.000e-01	450	5.915e-02
1196	5.883e+01	452	2.832e+00
1198	7.564e+00	453	2.921e+00
1199	5.556e+01	454	2.567e+00
1200	3.076e+01	455	4.412e-01
1202	3.066e+01	456	1.842e+00
1203	5.882e+01	457	2.625e+00
1204	2.195e+01	458	2.624e+00
1206	4.278e+01	459	1.140e+00
1209	3.207e+01	460	1.534e+00
1212	8.317e+00	461	1.048e+00
1217	4.240e+01	462	1.353e+00
1232	1.329e+01	463	1.818e+00
1235	2.600e+01	464	1.500e+00
1238	7.941e+01	465	1.209e-01
1254	5.850e+01	470	5.943e-01
1256	1.281e+02	471	8.869e-01
1258	2.620e+01	472	2.331e-02
1259	1.530e+01	475	7.572e-01
1262	4.102e+00	476	5.277e-01
1263	6.911e+02	477	9.083e-01
1264	2.740e+02	478	7.660e-01
1266	3.741e+02	479	2.355e-01
1268	1.897e+02	480	7.667e+00
1271	2.600e+00	483	5.824e-01
1272	2.820e+01	486	5.662e+00
1273	1.988e+02	488	5.435e-01
1274	9.060e+00	494	4.933e-03
1275	5.300e+00	499	6.444e-01
1278	5.690e+01	500	1.099e+01
1280	3.668e+02	502	6.387e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1284	6.990e+01	503	3.671e+00
1285	3.144e+01	504	1.763e+00
1286	4.531e+01	505	2.408e+00
1288	8.004e+00	506	2.006e+00
1292	2.420e+01	507	1.148e+00
1294	7.523e+00	508	5.615e-01
1296	4.939e+01	510	1.165e-01
1302	3.869e-01	511	8.725e-02
1305	1.763e+02	512	1.899e-01
1308	1.376e+02	513	3.318e-01
1317	6.880e+01	514	2.257e-01
1319	7.243e+01	515	4.218e-02
1320	2.499e+01	517	3.935e-02
1321	1.161e+02	518	6.955e-02
1322	1.325e+02	519	2.401e-01
1325	2.762e+01	520	3.162e-01
1328	8.427e+00	521	2.274e-01
1333	6.000e+00	524	4.656e-01
1334	2.831e-01	525	3.371e-01
1335	6.493e+00	526	2.410e-02
1339	1.597e+01	527	6.315e-01
1341	5.334e+00	528	5.223e-01
1343	1.427e+01	529	4.764e-01
1344	1.505e+01	530	2.270e-01
1345	2.245e+01	531	2.575e-01
1346	2.323e+01	533	4.406e-01
1349	2.088e+01	534	4.062e-01
1350	2.834e+01	535	4.326e-01
1355	7.731e+00	537	6.326e-01
1356	8.696e-01	538	8.648e-02
1358	1.462e+01	541	6.026e-01
1362	2.969e+00	542	1.801e-01
1367	5.345e+00	544	8.577e-01
1370	1.289e+01	545	8.282e-01
1371	2.512e+01	546	8.541e-01
1373	2.418e+00	550	1.013e+00
1391	5.037e+00	552	1.005e-02
1396	7.505e+01	553	9.981e+00
1397	1.882e+01	555	2.018e+01
1399	3.083e+01	556	1.993e+01
1400	4.060e+00	557	1.018e+01
1403	6.915e+01	559	9.828e-03
1404	9.792e+01	560	6.694e-03
1405	2.072e+01	564	3.617e-01
1408	4.591e+01	566	5.904e-01
1409	9.000e+00	567	5.005e-01
1410	5.211e+01	568	5.433e-01
1415	2.499e+02	570	2.199e-02
1416	4.109e+00	575	2.888e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1417	1.607e+01	577	7.964e-02
1418	1.337e+01	579	1.063e+00
1420	6.992e+00	580	1.095e+00
1422	8.824e+00	581	8.960e-01
1423	2.667e+00	584	8.138e-01
1424	5.992e+00	585	3.913e-01
1426	1.560e+01	586	1.171e-01
1427	6.361e+01	588	1.887e+01
1429	3.300e+00	589	1.690e+00
1432	1.311e+00	590	2.661e+00
1436	2.465e+02	591	1.274e+00
1440	3.184e+01	592	9.029e-01
1441	4.700e+02	593	5.427e-01
1445	3.323e+01	595	7.539e-01
1448	5.765e+01	596	2.112e-01
1450	2.531e+02	597	3.365e-01
1452	1.000e+02	598	2.933e+00
1457	3.529e+02	602	2.250e-01
1460	4.975e+01	604	2.249e-01
1461	1.171e+02	605	2.061e-01
1463	1.238e+02	606	4.122e-01
1464	3.040e+01	608	4.082e-01
1468	2.001e+01	610	5.375e-01
1473	9.002e+01	612	5.375e-01
1476	4.940e+00	613	2.683e+00
1479	1.026e+00	614	3.084e+00
1480	8.281e+00	617	5.439e+00
1485	2.716e+01	618	1.699e+01
1488	1.843e+01	619	1.306e+01
1489	1.295e+00	621	1.708e-01
1490	8.867e-01	623	5.973e-02
1492	1.032e+01	625	1.082e-01
1495	3.214e+01	626	3.015e-01
1496	2.103e+01	631	9.728e-02
1499	6.752e+01	634	8.940e-02
1502	3.115e+01	635	2.313e-01
1503	1.851e+01	641	1.493e-01
1506	1.072e+00	643	2.037e-02
1507	9.915e+00	644	3.923e+00
1517	7.496e+00	645	3.595e+00
1519	2.819e+01	646	3.698e-01
1526	2.877e+01	647	3.698e-01
1527	1.342e+01	648	6.967e-01
1543	7.660e+01	650	3.680e-01
1545	2.200e+01	654	3.295e+00
1546	4.104e+01	656	5.150e+00
1547	1.034e+02	657	4.664e+00

3 Variáveis Duais (Multiplicadores de Lagrange - Valores λ 0)

Tabela 2: Variáveis duais (y) com valores λ 0 do problema D2Q06C

Coordenada y	Valor y (Dual)
41	1.279e+00
47	5.408e-02
98	1.029e+01
99	8.736e+00
100	1.019e+01
101	1.029e+01
102	1.031e+01
106	7.699e+00
107	9.784e+00
108	1.266e+01
109	1.266e+01
110	1.247e+01
111	1.266e+01
112	8.499e+00
113	8.466e+00
114	1.184e+01
115	1.176e+01
117	1.268e+01
131	8.154e+00
132	9.812e+00
134	1.320e+01
135	1.026e+01
136	1.054e+01
137	9.487e+00
139	1.002e+01
140	9.859e+00
141	9.926e+00
158	1.049e+01
159	1.099e+01
160	1.123e+01
161	1.188e+01
162	1.148e+01
163	1.319e+01
164	1.323e+01
165	1.101e+01
166	1.099e+01
167	1.322e+01
168	1.382e+01
169	1.398e+01
170	1.382e+01
171	1.186e+01

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Coordenada y	Valor y (Dual)
172	1.162e+01
173	1.087e+01
174	1.193e+01
175	1.067e+01
176	1.324e+01
177	2.858e+00
178	1.058e+01
193	4.310e-02
206	9.471e+00
207	1.153e+01
208	1.132e+01
209	1.116e+01
210	1.113e+01
211	1.148e+01
212	1.070e+01
213	1.023e+01
214	1.411e+01
215	1.109e+01
216	1.088e+01
217	1.101e+01
218	1.066e+01
227	8.426e+00
228	1.999e+01
229	9.261e+00
230	1.151e+01
231	1.151e+01
232	1.151e+01
233	1.680e+01
234	9.261e+00
235	9.261e+00
236	9.261e+00
237	1.684e+01
238	1.573e+01
239	1.497e+01
240	2.351e+01
241	2.485e+01
242	1.669e+01
243	8.426e+00
244	1.108e+01
245	1.028e+01
246	1.056e+01
247	1.287e+01
248	1.686e+01
249	3.518e+01
250	4.165e-01
270	7.591e+00
271	1.110e+01
272	1.103e+01
273	1.079e+01
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Coordenada y	Valor y (Dual)
274	1.061e+01
275	1.017e+01
276	1.046e+01
277	1.009e+01
278	9.940e+00
279	1.245e+01
280	1.061e+01
281	1.090e+01
282	1.082e+01
294	1.327e+01
295	1.309e+01
296	1.145e+01
297	1.208e+01
298	1.140e+01
299	9.889e+00
300	1.501e+01
301	1.503e+01
302	1.145e+01
303	1.145e+01
304	1.577e+01
305	1.549e+01
306	1.297e+01
307	1.314e+01
308	1.513e+01
309	1.514e+01
310	1.127e+01
311	1.101e+01
312	1.069e+01
313	1.071e+01
314	1.059e+01
315	1.061e+01
316	9.200e+00
318	4.819e+00
319	3.494e-01
320	3.152e-01
334	1.799e+00
341	9.389e+00
342	1.149e+01
343	1.166e+01
344	1.009e+01
345	9.990e+00
346	1.028e+01
347	1.055e+01
348	1.055e+01
349	1.047e+01
350	1.021e+01
355	9.226e+00
356	1.077e+01
357	1.009e+01
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Coordenada y	Valor y (Dual)
358	8.856e+00
359	1.380e+01
360	1.338e+01
361	1.077e+01
362	1.077e+01
363	1.135e+01
364	1.444e+01
365	1.134e+01
366	1.134e+01
367	1.191e+01
368	9.361e+00
369	9.074e+00
371	1.044e+01
383	1.215e+00
392	9.203e+00
393	1.000e+01
394	1.111e+01
395	1.063e+01
396	1.039e+01
397	9.788e+00
398	9.799e+00
399	1.116e+01
400	1.083e+01
401	1.073e+01
409	1.160e+01
410	1.333e+01
411	1.950e+01
412	1.301e+01
413	1.386e+01
414	1.146e+01
415	1.608e+01
416	1.160e+01
417	1.414e+01
418	1.366e+01
419	1.409e+01
420	9.979e+00
421	9.979e+00
422	9.121e+00
423	1.370e+01
424	4.831e+00
425	3.586e+01
426	3.697e-01
427	7.509e-01
437	5.532e-03
445	8.990e+00
446	1.068e+01
447	1.063e+01
448	1.105e+01
449	1.032e+01
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Coordenada y	Valor y (Dual)
450	1.071e+01
451	9.833e+00
452	1.088e+01
453	1.110e+01
454	1.107e+01
455	1.074e+01
456	1.075e+01
476	9.062e+00
477	8.397e+00
478	1.348e+01
479	1.330e+01
480	1.126e+01
481	7.028e+00
482	8.604e+00
483	8.604e+00
484	1.005e+01
485	1.043e+01
486	1.331e+01
487	1.357e+01
488	1.315e+01
489	1.101e+01
490	8.604e+00
491	1.331e+01
492	1.331e+01
493	1.086e+01
494	1.042e+01
495	1.040e+01
496	1.183e+01
497	1.208e+01
498	1.183e+01
499	9.014e+00
500	7.795e+00
501	1.180e+01
502	1.369e+01
504	3.482e+00
514	2.269e-02
529	7.164e+00
530	1.120e+01
531	1.115e+01
532	1.090e+01
533	1.023e+01
534	1.102e+01
539	2.333e+01
540	1.593e+01
541	1.517e+01
542	6.015e+00
543	1.009e+01
544	1.024e+01
545	1.575e+01
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Coordenada y	Valor y (Dual)
546	6.064e+00
547	6.015e+00
548	1.273e+01
549	1.537e+01
550	1.183e+01
551	1.120e+01
552	8.644e+00
553	2.923e+00
554	1.710e+01
555	1.107e+00
556	4.293e+00
575	5.468e+00
576	1.146e+01
577	8.936e+00
578	1.056e+01
579	1.163e+01
580	1.061e+01
581	9.951e+00
582	1.024e+01
583	1.052e+01
584	1.032e+01
585	9.832e+00
586	1.047e+01
587	1.132e+01
588	1.045e+01
589	1.055e+01
590	1.036e+01
599	1.071e+01
600	9.337e+00
601	1.539e+01
602	1.258e+01
603	1.259e+01
604	1.259e+01
605	2.625e+01
606	9.337e+00
607	1.225e+01
608	1.253e+01
609	1.260e+01
610	1.260e+01
611	1.253e+01
612	1.289e+01
613	1.263e+01
614	1.249e+01
615	1.258e+01
617	9.495e+00
639	7.981e+00
640	1.064e+01
641	1.069e+01
642	9.255e+00

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Coordenada y	Valor y (Dual)
643	9.646e+00
644	9.933e+00
645	1.022e+01
646	1.002e+01
647	1.024e+01
648	1.011e+01
649	9.747e+00
650	9.603e+00
651	9.363e+00
652	9.699e+00
659	8.469e+00
660	8.512e+00
661	1.003e+01
662	1.153e+01
663	1.034e+01
664	1.296e+01
665	1.296e+01
666	2.625e+01
667	9.859e+00
668	1.186e+01
669	1.286e+01
670	1.312e+01
671	1.250e+01
672	1.250e+01
673	1.171e+01
674	1.202e+01
675	1.186e+01
676	1.307e+01
677	1.133e+01
681	1.082e+01
694	4.491e-01
698	2.307e+00
702	1.666e+01
704	8.573e+00
705	1.051e+01
706	1.041e+01
707	1.017e+01
708	1.052e+01
715	1.291e+01
716	1.341e+01
717	1.174e+01
718	1.119e+01
719	8.577e+00
720	1.381e+01
721	1.352e+01
734	4.341e+00
739	1.076e+01
740	1.092e+01
741	1.037e+01
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Coordenada y	Valor y (Dual)
742	1.039e+01
743	1.091e+01
744	1.101e+01
749	1.220e+01
750	1.276e+01
751	9.590e+00
752	1.280e+01
753	1.015e+01
754	1.691e+01
755	8.076e+00
756	8.076e+00
757	1.818e+01
758	1.316e+01
759	1.305e+01
761	3.780e-01
780	1.017e+01
781	6.570e+00
782	1.102e+01
783	1.141e+01
784	1.140e+01
785	1.047e+01
786	1.049e+01
787	1.105e+01
788	1.114e+01
795	1.468e+01
796	6.657e+00
797	1.474e+01
798	8.170e+00
799	5.943e+00
800	8.064e+00
801	8.194e+00
802	7.938e+00
803	1.299e+01
804	1.286e+01
814	1.569e+00
820	2.423e+02
823	1.646e+00
825	1.223e+01
826	1.056e+01
827	1.048e+01
828	1.079e+01
829	1.045e+01
830	1.057e+01
835	1.178e+01
836	1.632e+01
837	6.561e+00
838	1.682e+01
839	1.077e+01
840	9.638e+00
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Coordenada y	Valor y (Dual)
841	1.623e+01
842	1.391e+01
843	1.187e+01
855	3.664e+01
861	3.616e+01
866	9.816e+00
867	1.322e+01
868	1.240e+01
869	1.099e+01
870	1.106e+01
871	1.119e+01
872	1.446e+01
873	1.277e+01
874	1.382e+01
875	1.324e+01
876	1.114e+01
877	1.357e+01
878	1.101e+01
879	8.604e+00
880	1.470e+01
881	1.462e+01
882	1.317e+01
883	1.235e+01
884	1.230e+01
885	1.214e+01
886	1.227e+01
887	1.203e+01
888	1.175e+01
889	1.473e+01
890	1.469e+01
891	1.439e+01
892	1.469e+01
894	1.377e+01
895	1.331e+01
896	1.410e+01
897	1.284e+01
898	1.164e+01
899	1.140e+01
900	1.145e+01
901	1.134e+01
902	1.404e+01
903	1.469e+01
904	1.326e+01
905	1.396e+01
906	1.374e+01
907	1.374e+01
908	3.901e+00
909	3.635e+00
910	4.619e+00

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Coordenada y	Valor y (Dual)
911	4.112e+00
912	3.679e+00
913	4.058e+00
914	4.148e+00
915	4.100e+00
916	4.134e+00
930	4.253e+00
931	5.259e+00
932	4.923e+00
933	4.297e+00
934	4.297e+00
935	4.347e+00
936	4.297e+00
937	4.780e+00
938	4.303e+00
939	4.303e+00
940	4.405e+00
941	4.405e+00
942	4.405e+00
943	4.257e+00
944	4.252e+00
945	3.903e+00
946	4.305e+00
947	4.305e+00
948	4.316e+00
949	4.455e+00
950	4.135e+00
952	4.225e+00
974	4.068e+00
975	1.046e+01
976	1.003e+01
977	9.764e+00
978	1.037e+01
979	9.269e+00
980	9.292e+00
981	1.192e+01
982	9.609e+00
983	1.084e+01
984	9.422e+00
985	9.519e+00
1014	1.354e+01
1015	1.234e+01
1016	1.217e+01
1017	1.063e+01
1018	1.017e+01
1019	1.017e+01
1020	1.017e+01
1021	1.223e+01
1022	9.109e+00
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Coordenada y	Valor y (Dual)
1023	1.063e+01
1024	1.063e+01
1025	1.113e+01
1026	1.230e+01
1027	1.063e+01
1028	1.064e+01
1029	1.063e+01
1030	1.063e+01
1031	1.127e+01
1032	1.127e+01
1033	1.117e+01
1034	1.047e+01
1035	1.095e+01
1036	1.237e+01
1054	1.312e+00
1062	9.661e+00
1063	1.028e+01
1064	1.075e+01